

Key Definitions:

1. Kernel (N_A):

- The kernel of a matrix A is the set of all vectors x such that:

$$Ax = 0,$$

where 0 is the zero vector.

- In simpler terms, it is the set of all solutions to the equation $Ax = 0$.

2. Nullity ($v(A)$):

- The nullity of A is the **dimension of the kernel**. It tells us how many independent solutions exist in N_A .

3. Key Tools:

- To find N_A , we solve the equation $Ax = 0$ using **row reduction** to get the matrix in row-echelon form.
- Nullity is calculated as:

$$v(A) = n - r,$$

where n is the number of columns of A , and r is the rank (number of non-zero rows in the row-echelon form).

Part (a) (i):

Let:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \end{bmatrix}.$$

Step 1: Solve $Ax = 0$:

- Write $Ax = 0$:

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

2. Expand into equations:

$$1 \cdot x_1 + 2 \cdot x_2 - 1 \cdot x_3 = 0,$$

$$2 \cdot x_1 - 1 \cdot x_2 + 3 \cdot x_3 = 0.$$

Step 2: Row Reduce the Matrix:

1. Write A as an augmented matrix:

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 2 & -1 & 3 & 0 \end{bmatrix}.$$

2. Perform row reduction:

- Subtract 2 times the first row from the second row:

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & -5 & 5 & 0 \end{bmatrix}.$$

3. Simplify the second row:

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix}.$$

4. Eliminate the second entry in the first row: Subtract 2 times the second row from the first row:

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix}.$$

Step 3: Write the Solution:

1. From the row-reduced matrix:

$$x_1 + x_3 = 0, \quad x_2 - x_3 = 0.$$

2. Express x_1 and x_2 in terms of x_3 :

$$x_1 = -x_3, \quad x_2 = x_3.$$

3. Write the solution as:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}.$$

4. Kernel (N_A):

$$N_A = \text{Span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Step 4: Find Nullity ($v(A)$):

- $n = 3$ (number of columns),
- $r = 2$ (number of non-zero rows in row-echelon form),
- Nullity:

$$v(A) = n - r = 3 - 2 = 1.$$

Part (b) (ii):

Let:

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & -2 & 6 \\ -6 & 3 & -9 \end{bmatrix}.$$

Step 1: Solve $Ax = 0$:

1. Write $Ax = 0$:

$$\begin{bmatrix} 2 & -1 & 3 \\ 4 & -2 & 6 \\ -6 & 3 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Step 2: Row Reduce the Matrix:

1. Write A as an augmented matrix:

$$\begin{bmatrix} 2 & -1 & 3 & 0 \\ 4 & -2 & 6 & 0 \\ -6 & 3 & -9 & 0 \end{bmatrix}.$$

2. Perform row reduction:

- Divide the first row by 2:

$$\begin{bmatrix} 1 & -0.5 & 1.5 & 0 \\ 4 & -2 & 6 & 0 \\ -6 & 3 & -9 & 0 \end{bmatrix}.$$

- Subtract 4 times the first row from the second row and 6 times the first row from the third row:

$$\begin{bmatrix} 1 & -0.5 & 1.5 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Step 3: Write the Solution:

1. From the row-reduced matrix:

$$x_1 - 0.5x_2 + 1.5x_3 = 0.$$

2. Express x_1 in terms of x_2 and x_3 :

$$x_1 = 0.5x_2 - 1.5x_3.$$

3. Write the solution as:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 0.5 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1.5 \\ 0 \\ 1 \end{bmatrix}.$$

4. Kernel (N_A):

$$N_A = \text{Span} \left\{ \begin{bmatrix} 0.5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1.5 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Step 4: Find Nullity ($v(A)$):

- $n = 3$ (number of columns),
- $r = 1$ (number of non-zero rows in row-echelon form),
- Nullity:

$$v(A) = n - r = 3 - 1 = 2.$$

Final Results:

1. (a) (i):

- Kernel (N_A):

$$\text{Span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

- Nullity ($v(A)$): 1.

2. (b) (ii):

- Kernel (N_A):

$$\text{Span} \left\{ \begin{bmatrix} 0.5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1.5 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

- Nullity ($v(A)$): 2.